Module 2: Advanced Critical Software Engineering Course

Prerequisites: Critical Software Engineering

Lessons

No	Lesson title	Description	Video support (min)	Study/lab (hrs.)
1	Configuration Management Tools	Safety Standards require a rigorous Configuration Management Approach. The lesson will show how to keep SCADE projects under configuration using standard tools like GitHub.	15	2 Git lab
2	Verification of Critical Software with SCADE	The verification is a crucial phase in the development of Safety Critical Applications, as it ensures that the software operates safely, reliably, and adheres to the stringent safety standards. The objectives and strategies will be discussed in this lesson, emphasizing that SCADE allows simulation and testing of models before code generation, ensuring the design meets the software requirements. Also, through the SCADE Suite Requirements Management Gateway, requirements can be linked to model elements, allowing full traceability.	15	4 SCADE Test lab
3	Change Management	The lesson addresses change management in safety-critical software, focusing on using model differencing with SCADE to identify and correct regressions in system behavior. It emphasizes maintaining traceability, ensuring compliance, and validating modifications to uphold system safety and reliability.	15	2
4	Continuous Integration/ Continuous Development	The lesson introduces CI/CD practices tailored for safety-critical software development, integrating SCADE model workflows with PyAnsys scripting to automate testing and validation. It highlights how continuous integration and delivery can enhance reliability, traceability, and compliance while maintaining rigorous certification standards.	15	4